

TRACE

Tracking, Recognition, Analytics & City-wide Traffic Enforcement

Project Requirement Document — Revised v1.1

Revision: Section 7.2 team-composition wording only. All functional requirements, NFRs, scope, milestones and acceptance criteria remain unchanged.

Problem Statement ID: 26127 | Organization: Bharat Electronics Limited | Category: Software | Theme: Transportation & Logistics

1. Introduction

1.1 Purpose

This PRD defines what TRACE must achieve to satisfy Problem Statement 26127 and the team's quality bar of a complete, working, flaw-free system. It is the source document for the Technical Requirement Document, App/Website Flow, UI/UX Brief, Backend Schema and Implementation Plan.

1.2 Project Background

Modern cities operate large networks of CCTV and ANPR cameras, but these networks typically function as isolated silos. TRACE addresses this by linking observations across space and time and extracting city-level traffic patterns.

1.3 Guiding Principle

TRACE is a spatio-temporal vehicle intelligence system structured around five layers: Layer 1 Perception; Layer 2 Identity; Layer 3 Spatial-Temporal Reasoning; Layer 4 Network Intelligence; Layer 5 Prediction & Anomaly Reasoning.

2. Project Objectives

- OBJ-1: Deliver an ANPR/OCR engine with greater than 90% recognition accuracy across varied real-world conditions.
- OBJ-2: Reconstruct the complete city-wide trajectory of any queried vehicle plate with timestamps, direction and route on a GIS map.
- OBJ-3: Compute and visualize macro traffic analytics including density, origin-destination patterns, congestion bottlenecks and heatmaps.
- OBJ-4: Flag blacklisted vehicles and anomalous routes in real time through an alert system.
- OBJ-5: Differentiate TRACE through multi-modal identity fusion, road-graph-constrained reasoning and self-validating anomaly detection.
- OBJ-6: Deliver a working, end-to-end, demo-ready prototype covering every functionality above across six milestones.

3. Scope

3.1 In-Scope — Core Build

- Vehicle detection, tracking and plate localization from live or simulated/recorded camera feeds.
- OCR with confidence scoring and temporal evidence aggregation.
- Multi-modal identity fusion using plate text, OCR confidence, vehicle type and colour.
- Road-graph model of camera nodes and connecting road segments.
- Query-based trajectory reconstruction on a GIS map.
- Impossible-journey anomaly detection using road-graph travel-time constraints.
- City Traffic Analytics Dashboard with heatmaps, density and route statistics.
- Blacklist-based real-time alerting.

3.2 In-Scope — Narrated / Lightweight Implementation

- Congestion prediction using a heuristic or lightweight time-series projection.

- Camera reliability profiling using a static configuration table.

3.3 Out of Scope — Production Roadmap

- Kafka/Redpanda event streaming and Apache Flink stream processing.
- Kubernetes orchestration and NVIDIA DeepStream/TensorRT edge inference deployment.
- Neo4j-based knowledge graph.
- Counterfactual what-if traffic simulation.
- Full enterprise privacy pipeline; only minimal access control is implemented in the prototype.

4. Stakeholders

Stakeholder	Role	Primary interest
Bharat Electronics Limited	Problem statement owner / evaluator	Deployable, credible city-scale traffic intelligence platform
City Traffic Authority	Simulated end user	Fast, reliable vehicle tracing and traffic oversight
Traffic Analyst	Simulated end user	Accurate macro traffic patterns and forecasts
Project Team	Builders	Complete, working, well-documented submission
Evaluators / Judges	Assessors	Technical depth, novelty, feasibility and working demo

5. Functional Requirements

5.1 Layer 1 — Perception

ID FR-PER-	Requirement
01 FR-	Detect vehicles and license plates in each camera frame using an object detection model.
PER-02	Track a vehicle across consecutive frames within a single camera view to avoid duplicate counting.
FR-PER-03	Extract plate text via OCR and attach a confidence score to each character/plate read.
FR-PER-04	Aggregate multiple OCR reads of the same vehicle into a single higher-confidence plate result.
FR-PER-05	Achieve greater than 90% plate recognition accuracy across varied lighting, angle, blur and plate-condition scenarios.

5.2 Layer 2 — Identity Fusion

ID FR-	Requirement
ID-01	Record plate text, vehicle type and colour at each observation.
FR-ID-02	Compute a composite identity-match confidence score using plate similarity, OCR confidence and attribute similarity.
FR-ID-03	Propose candidate matches when OCR is partial or low-confidence using non-plate attributes as supporting evidence.
	Expose identity-match confidence wherever a match is displayed.
FR-ID-04	

5.3 Layer 3 — Spatial-Temporal Reasoning

ID	Requirement
FR-STR-01	Maintain a road graph where camera locations are nodes and road segments are edges with distance and expected travel-time attributes.
FR-STR-02	Restrict candidate matches to graph-reachable cameras within the observed time gap.
FR-STR-03	Reconstruct a vehicle's full trajectory as an ordered sequence of camera observations with timestamps.
FR-STR-04	Flag an impossible journey when implied travel speed exceeds a configurable plausible threshold.
FR-STR-05	Allow an operator to query a plate and receive its reconstructed trajectory on a GIS map.

5.4 Layer 4 — Network Intelligence

ID FR-ANL-	Requirement
01 FR-	Compute traffic density per camera node / road segment over configurable time windows.
ANL-02	Compute an origin-destination matrix from aggregated trajectory data.
FR-ANL-03	Identify congestion bottlenecks based on density and average speed thresholds.
FR-ANL-04	Render a real-time city-wide traffic heatmap on the GIS dashboard.
FR-ANL-05	Display average vehicle speed and route density per segment.

5.5 Layer 5 — Prediction & Anomaly Reasoning

ID	Requirement
FR-PRD-01	Produce a short-horizon congestion forecast (e.g. 15-30 minutes) per monitored segment using recent flow trends.
FR-PRD-02	Flag suspicious route anomalies including impossible journeys, duplicated-plate detections and camera-timestamp inconsistencies.
FR-PRD-03	Maintain a camera reliability profile used to weight evidence from less-reliable cameras lower during identity fusion.

5.6 Alert System

ID FR-ALT-	Requirement
01 FR-	Maintain a blacklist of vehicle plates of interest.
ALT-02	Generate a real-time alert when a blacklisted plate is recognized at any camera node, showing location and timestamp.
FR-ALT-03	Generate an alert when a Layer 5 anomaly is detected, distinguishing anomaly type.
FR-ALT-04	Log all generated alerts with timestamp, camera, plate and alert type.

6. Non-Functional Requirements

Category	ID	Requirement
Accuracy	NFR-01	OCR/ANPR recognition accuracy shall exceed 90% on the evaluation dataset.
Performance	NFR-02	Trajectory queries shall return within an acceptable interactive response time on the demo dataset.
Scalability	NFR-03	Additional camera nodes can be added without redesigning the schema.
Usability	NFR-04	Dashboard shall present trajectory, heatmap and alert views through a single consistent navigation structure.
Reliability	NFR-05	System shall degrade gracefully when a camera feed is unavailable, rather than failing the entire pipeline.
Security & Privacy	NFR-06	Access to trajectory and identity data shall be role-restricted; production encryption/retention path documented. Every identity match and anomaly flag shall have a confidence score or reason code.
Explainability	NFR-07	Each of the five layers shall be implemented as a separable module with a defined interface.
Maintainability	NFR-08	

7. Assumptions and Constraints

7.1 Assumptions

- Livecamera feeds from an actual city ANPR network will not be available; recorded or simulated multi-camera footage with known ground-truth plates will be used.
- Road-graph distances and expected travel times can be approximated using map data for the demo area.
- The demo dataset will contain a manageable number of camera nodes.

7.2 Constraints — REVISED

- The project must be completed within the timeline defined in the Implementation Plan, by the team constituted for execution — originally scoped as six roles, executed as three (P1 Frontend, P2 Backend + ML, P3 Data Collection & Support) per the Team Roles & Milestone Contribution Guide v1.1 and Implementation Plan v1.3.

- The technology stack is deliberately limited to tools the team can realistically operate under deadline pressure.
- Kafka, Kubernetes, Neo4j, and edge-hardware deployment are constrained to design documentation only.

8. Dependencies and Traceability

The PRD remains the source document for the Technical Requirement Document, App/Website Flow, UI/UX Brief, Backend Schema and Implementation Plan. The v1.3 Implementation Plan and Team Roles Guide define current execution ownership.

9. Success and Acceptance Criteria

- All FR-xxx requirements are implemented and demonstrable in a single end-to-end run.
- OCR/ANPR accuracy exceeds 90% on the evaluation dataset.
- A queried plate returns a correct chronological GIS trajectory including an impossible-journey flag.
- Analytics displays heatmap, density and OD-pattern views from the same trajectory dataset.
- A blacklisted plate triggers a visible real-time alert during the live demo.
- The accompanying documents remain mutually consistent.

10. Glossary

Term	Definition
ANPR	Automatic Number Plate Recognition.
OCR	Optical Character Recognition used to read plate regions.
Identity Fusion	Combining plate text, OCR confidence and vehicle attributes into one confidence-scored identity.
Road Graph	City representation where cameras are nodes and road segments are edges carrying distance/travel-time data.
Impossible Journey	Flagged anomaly where observations imply physically implausible speed.
OD Matrix	Origin-Destination matrix summarizing trip volumes between zones/camera nodes.

Term	Definition
Layer Model	Five-layer intelligence framework: Perception, Identity, Spatial-Temporal Reasoning, Network Intelligence, Prediction & Anomaly.